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Nonlinear Effects of Digital Factor Allocation on Regional Coordinated Development: Evidence from a Provincial Panel Threshold Model in China

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Abstract

China's digital economy has grown rapidly, but whether digital factor allocation promotes regional coordinated development remains an open question. This study builds a digital factor allocation (DFA) index covering digital infrastructure, digital industry, digital finance, and digital human capital, and a regional coordinated development (RCD) index covering economic, social, and urban-rural coordination, for 30 Chinese provinces from 2013 to 2021. Using entropy weights and a Hansen panel threshold model, we estimate the nonlinear relationship between the two indices. DFA is positively associated with RCD: the baseline two-way fixed-effects estimate is 0.241 ($p < 0.05$). When per capita GDP exceeds the estimated threshold of 10.589 (about CNY 39,700), the marginal association rises from 0.506 to 0.677; a financial development threshold indicates declining returns at very high levels of financial depth. The association is strongest in western provinces, and digital finance carries most of it. Both indices are strongly spatially clustered. Spatial specifications leave the threshold location unchanged but shrink the regime difference, so we read the nonlinearity as a development gradient rather than a sharp break; the positive association survives every specification. Digital factor policies should therefore be sequenced with economic development, tailored to regional absorptive capacity, and paired with complementary reforms in advanced regions.

Keywords: digital factor allocation; regional coordinated development; threshold effect; absorptive capacity; digital finance; China

1. Introduction

China has built one of the world's largest digital economies. Broadband, mobile payments, e-commerce, and cloud-based public services have spread from coastal cities into inland and rural areas, changing how firms organize production, how households access finance, and how governments deliver services (Goldfarb & Tucker, 2019). The gains are uneven. Per capita income, public-service capacity, and digital infrastructure differ sharply across provinces, and the gap between leading and lagging regions is a central concern of regional policy (Kanbur & Zhang, 2005; World Bank, 2009).

Digital factor allocation (DFA)—the combined availability and application of digital infrastructure, digital industry, digital finance, and digital human capital—is a potential

Academic Editor: Firstname Lastname

Received:
Accepted:
Published:

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lever for regional coordinated development (RCD). Digital infrastructure lowers the cost of market access; digital finance extends payment and credit to households and firms that were previously excluded; digital industry creates new tasks and occupations; and digital human capital determines whether these opportunities are absorbed locally (Czernich et al., 2011; Hjort & Poulsen, 2019; Jack & Suri, 2014; Liu et al., 2023). The same forces can widen gaps. Digital investment clusters where complementary skills, finance, and infrastructure already exist, platform markets reward scale, and lagging regions may gain access without the capability to convert it into income (Acemoglu & Restrepo, 2020; Akerman et al., 2015; Forman et al., 2012; van Dijk, 2005). Whether DFA promotes or undermines RCD is therefore an empirical question.

Existing work falls into three strands. Measurement studies build composite digital indices (Chen & Zhang, 2025; Peking University Digital Finance Research Center, 2021) that rank regions but do not distinguish infrastructure, industry, finance, and human capital. Effect studies document positive associations with growth, productivity, and inclusion, although the estimates vary with the level of development (Akerman et al., 2015; Czernich et al., 2011; Hjort & Poulsen, 2019; Liu et al., 2023). Threshold studies show that digitalization can have stage-dependent effects (Qi et al., 2025). Three gaps follow. Composite indices obscure which digital factor matters; RCD is often proxied by a single indicator rather than measured as a multidimensional outcome; and threshold evidence for China remains thin.

This study builds a four-dimension DFA index and a three-dimension RCD index for 30 provinces from 2013 to 2021, and applies a Hansen panel threshold model. It then asks which dimension carries the association, how the association varies across regions, and whether the estimates survive spatial dependence.

We make four contributions. First, we measure DFA as four separate factors and RCD as three coordination dimensions, so the analysis identifies which factor matters rather than averaging four heterogeneous components. Second, we test the nonlinearity with bootstrap and likelihood-ratio inference and subject it to spatial and sub-dimension checks. Third, we show that digital finance carries most of the association while industrial digitalization and digital human capital change sign across the threshold. Fourth, the results imply a differentiated sequencing of digital policy across regions.

The results are as follows. DFA is positively associated with RCD: the baseline two-way fixed-effects estimate is 0.241 ($p < 0.05$). The association rises with economic development. When per capita GDP exceeds the estimated threshold of 10.589 (about CNY 39,700), the marginal association rises from 0.506 to 0.677. A financial development threshold indicates declining returns at very high levels of financial depth. The association is strongest in western provinces. Both indices are spatially clustered; spatial specifications leave the threshold location unchanged but shrink the regime difference, while the positive association survives every specification. These findings are consistent with an absorptive-capacity interpretation.

Section 2 reviews the literature and states the hypotheses. Section 3 describes the data and methods. Section 4 reports the results, Section 5 discusses them, and Section 6 concludes.

2. Literature Review and Hypotheses

2.1. Digital Factor Allocation and Regional Coordinated Development

Regional coordinated development refers to the joint improvement of economic performance, social welfare, and urban-rural balance across regions, not simply the convergence of GDP per capita. It also requires that public services, income opportunities, and basic infrastructure be accessible in different locations and that the gains from growth be

shared between leading and lagging areas (Kanbur & Zhang, 2005; World Bank, 2009). In China, RCD has been pursued through fiscal transfers, infrastructure investment, poverty alleviation, and, more recently, digitalization.

DFA can affect RCD through four channels. Digital infrastructure reduces communication and transaction costs, allowing firms in lagging regions to reach customers and information that were previously available only in core regions (Czernich et al., 2011; Hjort & Poulsen, 2019). Digital finance extends payment, savings, credit, and insurance to households and small firms, improving risk sharing and relaxing liquidity constraints (Beck et al., 2007; Jack & Suri, 2014; Liu et al., 2023). Digital industry creates new tasks and raises productivity in services that are less dependent on physical distance (Acemoglu & Autor, 2011; Goldfarb & Tucker, 2019). Digital human capital determines whether local workers can use digital tools productively (Akerman et al., 2015; van Dijk, 2005).

This reasoning supports a positive relationship, but the empirical literature also cautions that the benefits are conditional. Forman et al. (2012) show that the internet's wage effects are concentrated in already prosperous areas, and Akerman et al. (2015) find that broadband complements skilled labor. Whether DFA has a uniform effect across Chinese regions is therefore an open question.

2.2. Absorptive Capacity and Threshold Effects

The idea that digitalization has stage-dependent effects comes from the absorptive-capacity literature. Digital technologies are general-purpose technologies whose effects depend on how they are combined with organizational change, human capital, and institutional quality (Brynjolfsson & McAfee, 2014; O'Mahony & Vecchi, 2005). In low-income regions with thin markets, digital inputs may be used for consumption rather than productive investment; in higher-income regions the same inputs can be combined with skills, finance, and infrastructure to generate larger coordination gains. This predicts that the marginal effect of DFA increases with economic development.

Hansen (1999) provides a fixed-effects panel threshold estimator that lets the slope of a regressor differ across regimes defined by an observed variable, searching for the threshold that minimizes the residual sum of squares and testing linearity by bootstrap. The approach has been extended to instrumental-variable and dynamic panels (Caner & Hansen, 2004; Seo & Shin, 2016) and is widely used on Chinese regional data. Qi et al. (2025), for example, find threshold effects of digital development, market efficiency, and technological progress on Chinese-style modernization, which suggests that the digital dividend depends on the broader development environment.

The threshold interpretation also connects to the digital divide. Access is only the first level; the second and third levels concern skills and the ability to convert access into material benefits (Norris, 2001; van Dijk, 2005; Warschauer, 2003). If the second and third divides persist in low-income regions, the effect of DFA on RCD may be weak there and stronger once complementary capabilities are in place.

2.3. Financial Development and Diminishing Returns

Digital finance is a core component of DFA, and its effect may depend on the depth of the financial system. Where financial development is limited, digital finance substitutes for missing branches and expands access to basic services. Where the system is already deep, it adds less at the extensive margin, may intensify competition, or may divert resources from productive investment. The relationship between financial depth and inclusion is therefore unlikely to be linear (Beck et al., 2007). Digital finance has expanded fastest in leading cities where financial services were already abundant, so its marginal contribution

to RCD may be larger where finance is scarce. This motivates a second threshold in financial development.

2.4. Regional Heterogeneity and Mechanisms

China's regions differ in initial digital endowments, industrial structure, and institutional capacity. The east has denser digital infrastructure, more high-skilled labor, and deeper financial markets; the centre and west start lower but may have more room for catch-up; the northeast carries a heavy-industry legacy and a slow digital transition. The absorptive-capacity argument predicts a stronger effect in the east, whereas the catch-up argument predicts a stronger effect in the west. The two can be distinguished empirically.

Three mechanisms are relevant. DFA can promote industrial upgrading by reducing information costs, expanding market access, and enabling new service activities (Chen & Zhang, 2025; Rodrik, 2016). It can deepen financial development by extending the coverage and use of digital financial services (Chen & Zhang, 2025; Liu et al., 2023). And it can raise technological investment by lowering the cost of innovation-related information and creating demand for digital skills (Akerman et al., 2015). The channel analysis in this study is exploratory.

2.5. Hypotheses

H1. Digital factor allocation is positively associated with regional coordinated development.

H2. The effect of DFA on RCD is nonlinear; the marginal effect is larger after economic development crosses a threshold, consistent with absorptive capacity.

H3. The effect of DFA varies with financial development and tends to diminish at very high levels of financial depth.

H4. The effect of DFA is regionally heterogeneous, with a stronger average effect in less-developed western provinces.

H5. DFA promotes RCD through industrial upgrading, financial deepening, and technological investment.

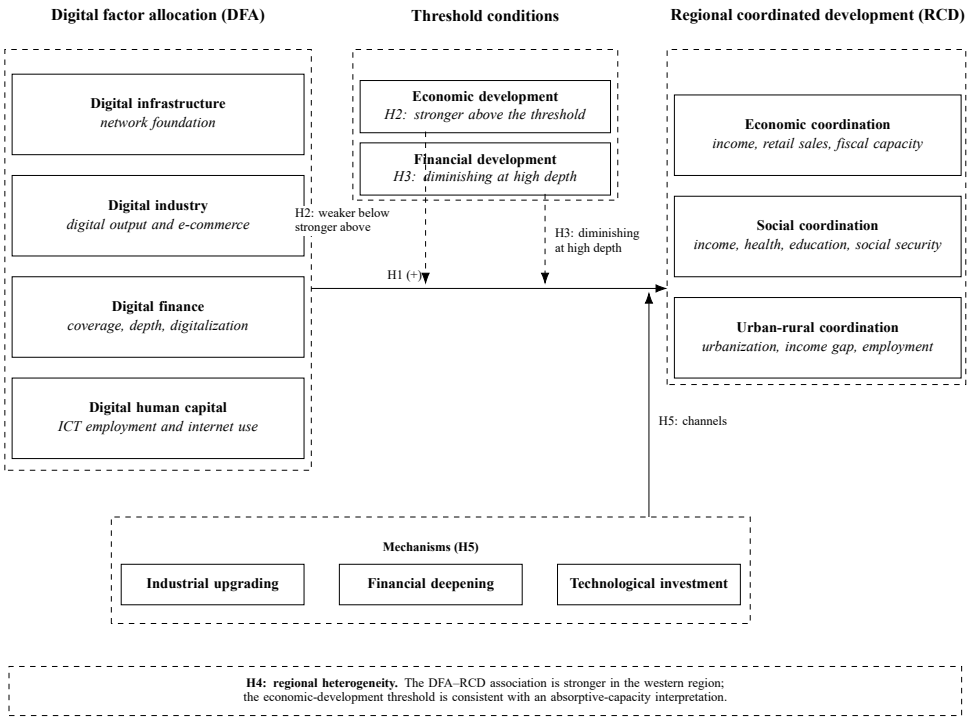


Figure 1. Research framework. DFA is measured across four dimensions and RCD across three dimensions. H1 tests the average association; H2 tests whether the DFA slope differs across economic-development regimes; H3 examines the financial-development threshold; H4 summarizes regional heterogeneity; H5 examines the channels.

3. Data, Variables, and Methods

3.1. Study Area and Data

The analysis covers 30 provincial-level units in mainland China from 2013 to 2021. Tibet, Hong Kong, Macao, and Taiwan are excluded because of data availability and comparability. The panel is balanced with 270 observations. The sample starts in 2013 because the National Bureau of Statistics introduced a unified urban-rural household survey that year, which makes the income series comparable over the period.

Provincial economic, social, infrastructure, and environmental indicators are taken from the *China Statistical Yearbook* (2014–2022), the *China City Statistical Yearbook* (2014–2022), the *China Environmental Statistics Yearbook*, and provincial yearbooks (National Bureau of Statistics of China, 2022a, 2022b). Digital financial inclusion sub-indices come from the Peking University Digital Financial Inclusion Index (Peking University Digital Finance Research Center, 2021). A small number of missing values were filled by linear interpolation within provinces. For urban and rural incomes, we checked the extracted values against the accounting identity linking total disposable income to urban and rural incomes and the urbanization rate, and corrected cells that violated it.

3.2. Digital Factor Allocation Index

DFA is measured with a four-dimension system (Table 1). Digital infrastructure captures the network foundations that determine whether digital services reach firms and households; digital industry captures the scale and market application of digital production; digital finance captures the coverage, intensity, and digitalization of financial services; and digital human capital captures the supply of workers who can use digital technologies.

The 15 indicators are selected for relevance to the dimension, availability across all provinces and years, and consistency of definitions. To prevent a few large provinces from

dominating the index, indicators are expressed as per capita, per GDP, or per area ratios. DFA assigns equal weights to the four dimensions and entropy weights to indicators within each dimension, so each factor category has the same ex ante importance while the data determine the weights inside it.

3.3. Regional Coordinated Development Index

RCD is measured with a three-dimension system (Table 1). Economic coordination uses GDP per capita, retail sales per capita, and fiscal self-sufficiency. Social coordination uses disposable income per capita, hospital beds per 10,000 people, higher-education students per 100,000 people, and the share of social security spending. Urban-rural coordination uses the urbanization rate, the urban-rural income ratio (negative), and the urban registered unemployment rate (negative).

The three dimensions receive equal weights and entropy weights are applied within each dimension. For robustness we also construct a five-dimension index that adds infrastructure and ecological coordination, a coupling coordination degree for the three core dimensions, and a global entropy-weighted index.

3.4. Entropy Weight Method

The entropy method assigns larger weights to indicators with greater dispersion. For indicator j and observation i, the raw value x_{ij} is winsorized at the 1st and 99th percentiles and normalized. For positive indicators,

$$z_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}, \tag{1}$$

and for negative indicators,

$$z_{ij} = \frac{\max_i x_{ij} - x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}. \tag{2}$$

A constant (10^{-4}) is added to avoid zeros. The normalized values become proportions $p_{ij} = z_{ij} / \sum_i z_{ij}$, and the entropy of indicator j is

$$e_j = -\frac{1}{\ln N} \sum_{i=1}^{N} p_{ij} \ln p_{ij}, \tag{3}$$

where N is the number of observations. The weight is

$$w_j = \frac{1 - e_j}{\sum_k (1 - e_k)}, \tag{4}$$

and the composite score is $S_i = \sum_j w_j z_{ij}$. Entropy weights are computed within each dimension and the dimension scores are averaged with equal weights, which prevents a dimension with many indicators from receiving disproportionate weight.

3.5. Baseline and Threshold Models

The baseline specification is a two-way fixed-effects model:

$$RCD_{it} = \alpha_0 + \alpha_1 DFA_{it} + \gamma' X_{it} + \mu_i + \delta_t + \varepsilon_{it}, \tag{5}$$

where i indexes provinces, t indexes years, μ_i and δ_t are province and year fixed effects, X_{it} is a vector of controls, and ε_{it} is clustered at the province level.

Table 1. Indicator systems for digital factor allocation (DFA) and regional coordinated development (RCD).

Dimension	Indicator	Definition and unit	Sign
Panel A: Digital factor allocation (DFA)			
Digital infrastructure	Internet access ports per capita	Internet broadband access ports / year-end population (ports per person)	+
	Optical cable density	Long-distance optical cable length / land area (km/km^2)	+
	Mobile phone penetration	Mobile phone subscribers per 100 people	+
	IPv4 addresses per capita	IPv4 addresses / year-end population	+
	Domain names per capita	Registered domain names / year-end population	+
Digital industry	Software revenue / GDP	Software business revenue / GDP (%)	+
	E-commerce sales / GDP	E-commerce transaction value / GDP (%)	+
	Firms with e-commerce	Share of enterprises with e-commerce transactions (%)	+
	Websites per 100 firms	Number of websites per 100 enterprises	+
	Information firms per capita	Information enterprises / year-end population	+
Digital finance	DFI coverage breadth	Peking University Digital Financial Inclusion Index sub-index	+
	DFI usage depth	Peking University Digital Financial Inclusion Index sub-index	+
	DFI digitization	Peking University Digital Financial Inclusion Index sub-index	+
Digital human capital	ICT employment share	Employment in information transmission, computer services, and software / total urban employment (%)	+
	Internet penetration	Internet users / year-end population (%)	+
Panel B: Regional coordinated development (RCD)			
Economic coordination	GDP per capita	Regional GDP / year-end population (CNY/person)	+
	Retail sales per capita	Retail sales of consumer goods / year-end population (CNY/person)	+
	Fiscal self-sufficiency	Local fiscal revenue / local fiscal expenditure (%)	+
Social coordination	Disposable income per capita	All residents' per capita disposable income (CNY/person)	+
	Hospital beds	Hospital beds per 10,000 people	+
	Higher-education students	Higher-education students per 100,000 people	+
	Social security spending	Social security and employment expenditure / fiscal expenditure (%)	+
Urban-rural coordination	Urbanization rate	Urban resident population / year-end population (%)	+
	Urban-rural income ratio	Urban disposable income / rural disposable income	−
	Urban unemployment rate	Registered urban unemployment rate (%)	−

Notes: All indicators are compiled from the *China Statistical Yearbook*, the *China City Statistical Yearbook*, provincial statistical yearbooks, and the Peking University Digital Financial Inclusion Index. DFA assigns equal weights to the four dimensions and entropy weights to indicators within each dimension. RCD assigns equal weights to the three dimensions and entropy weights to indicators within each dimension.

To test H2 and H3 we estimate a Hansen (1999) panel threshold model:

$$RCD_{it} = \beta_0 + \beta_1 DFA_{it} \cdot I(q_{it} \le \gamma) + \beta_2 DFA_{it} \cdot I(q_{it} > \gamma) + \gamma' X_{it} + \mu_i + \varepsilon_{it}, \quad (6)$$

where q_{it} is the threshold variable and $I(\cdot)$ is the indicator function. The threshold is estimated by grid search over the inner 80% of the distribution of q. The null of linearity is tested with a cluster wild bootstrap that generates $y^* = \hat{y} + \varepsilon^*$, where ε^* is the fitted residual multiplied by a province-level Rademacher sign, and re-estimates the threshold and the F statistic in each replication. The p-value is the share of bootstrap statistics above the observed one. We also report the likelihood-ratio confidence interval for γ using the critical value 7.35 (Hansen, 1999, 2000).

The threshold variable is economic development (log GDP per capita) for H2 and financial development for H3. We also estimate specifications using DFA, technological investment, and private economy vitality as threshold variables to check whether the nonlinearity is specific to economic development.

3.6. Controls and Descriptive Statistics

The controls capture local conditions that may affect RCD but are not part of the index. Government intervention is local expenditure over GDP; foreign direct investment is log realized FDI per capita; population density is log year-end population per square kilometer; and private economy vitality is private enterprises per 10,000 people. Together they cover the fiscal, external, spatial, and market dimensions of local conditions.

Table 2 reports descriptive statistics. DFA has a mean of 0.259 and a standard deviation of 0.179; RCD has a mean of 0.357 and a standard deviation of 0.148. The urban-rural income ratio averages 2.53 and ranges from 1.84 to 3.56.

Table 2. Descriptive statistics.

Variable	N	Mean	SD	Min	Median	Max
DFA index	270	0.259	0.179	0.007	0.223	1.000
RCD index	270	0.357	0.148	0.083	0.333	0.846
RCD5 index	270	0.414	0.117	0.137	0.406	0.805
RCD coupling coordination	270	0.555	0.180	0.051	0.550	1.000
Urban-rural income ratio	270	2.533	0.361	1.842	2.474	3.556
Government intervention	270	26.292	11.122	10.501	23.207	75.344
Log FDI per capita	270	18.762	1.250	15.540	18.402	21.650
Log population density	270	5.469	1.289	2.068	5.659	8.282
Private economy vitality	270	191.904	142.191	41.889	155.839	813.382
Financial development	270	7.604	3.072	3.241	6.892	19.634
Technological investment	270	2.204	1.511	0.539	1.559	6.757
Industrial structure upgrading	270	140.959	74.470	66.533	122.565	524.401

Notes: The sample covers 30 provinces from 2013 to 2021 (N = 270). DFA is the digital factor allocation index; RCD is the regional coordinated development index; RCD5 is the five-dimension alternative; RCD-CCD is the coupling coordination degree of the three core dimensions. See Table 1 for definitions.

3.7. Spatial Weights and Spatial Specifications

The spatial analysis uses the 30 provinces as units. The weight matrix W is a row-standardized rook contiguity matrix built from the provincial boundary file. Hainan has no land neighbour and is linked to Guangdong, the conventional treatment. As a robustness check we also use an inverse-squared-distance matrix from provincial centroids. Both have a zero diagonal, so W times a variable is the weighted average of the other provinces in the same year. Spatial autocorrelation is measured with the global Moran's I and with local

indicators (LISA), each with 999 random permutations; a province is classified as a local cluster when the pseudo p-value is below 0.10.

We estimate three spatial extensions of the threshold model. The first adds the spatial lag of the regressor,

$$\begin{aligned} RCD_{it} = \beta_1 DFA_{it} \cdot I(q_{it} \le \gamma) + \beta_2 DFA_{it} \cdot I(q_{it} > \gamma) \\ + \delta\,(W \cdot DFA)_{it} + \gamma' X_{it} + \mu_i + \varepsilon_{it}, \end{aligned} \tag{7}$$

which is not identified here: after the within transformation DFA and its spatial lag are correlated at 0.979, so the two coefficients cannot be separated with 30 provinces. We report it as a transparency check only.

The second is a spatial error model,

$$\begin{aligned} RCD_{it} &= \beta_1 DFA_{it} \cdot I(q_{it} \le \gamma) + \beta_2 DFA_{it} \cdot I(q_{it} > \gamma) + \gamma' X_{it} + \mu_i + u_{it}, \\ u_{it} &= \lambda \sum_j w_{ij} u_{jt} + e_{it}, \end{aligned} \tag{8}$$

and the third is a spatial autoregressive model,

$$\begin{aligned} RCD_{it} = \rho \sum_j w_{ij} RCD_{jt} + \beta_1 DFA_{it} \cdot I(q_{it} \le \gamma) \\ + \beta_2 DFA_{it} \cdot I(q_{it} > \gamma) + \gamma' X_{it} + \mu_i + \varepsilon_{it}. \end{aligned} \tag{9}$$

Both are estimated by maximum likelihood with province fixed effects. For a given spatial parameter the concentrated likelihood is maximized over γ by grid search, with the log-determinant evaluated from the eigenvalues of W. The threshold test is a cluster wild bootstrap conditional on the estimated spatial parameter.

4. Results

4.1. Spatial and Temporal Patterns

Figure 2 maps the two indices for 2013, 2017, and 2021. Both increased over the period: the national average DFA rose from 0.109 to 0.379 and the average RCD from 0.261 to 0.468. Every province recorded a higher value of both indices, and 29 of 30 provinces moved up at least one class in DFA and 27 in RCD.

The spatial distribution is persistent. The east has the highest values in every year; the northeast and central regions occupy the middle, with the northeast ahead in RCD throughout and the central region edging ahead in DFA from 2018; the west is last in most years. In 2021 the eastern averages were 0.548 for DFA and 0.587 for RCD, against 0.283 and 0.369 in the west.

Dispersion widened much more in DFA than in RCD. In 2013 the highest-DFA province had a value about 85 times the lowest; by 2021 the ratio had fallen to about four. Absolute dispersion nevertheless increased: the standard deviation of DFA rose from 0.129 to 0.177 and its range from 0.599 to 0.775. The RCD distribution moved far less, with the standard deviation going from 0.129 to 0.136 and the range from 0.569 to 0.564. Digital factors spread quickly in proportional terms and pulled apart in absolute terms, whereas coordination outcomes shifted more slowly. Province-by-year values for both indices are reported in Appendix B.

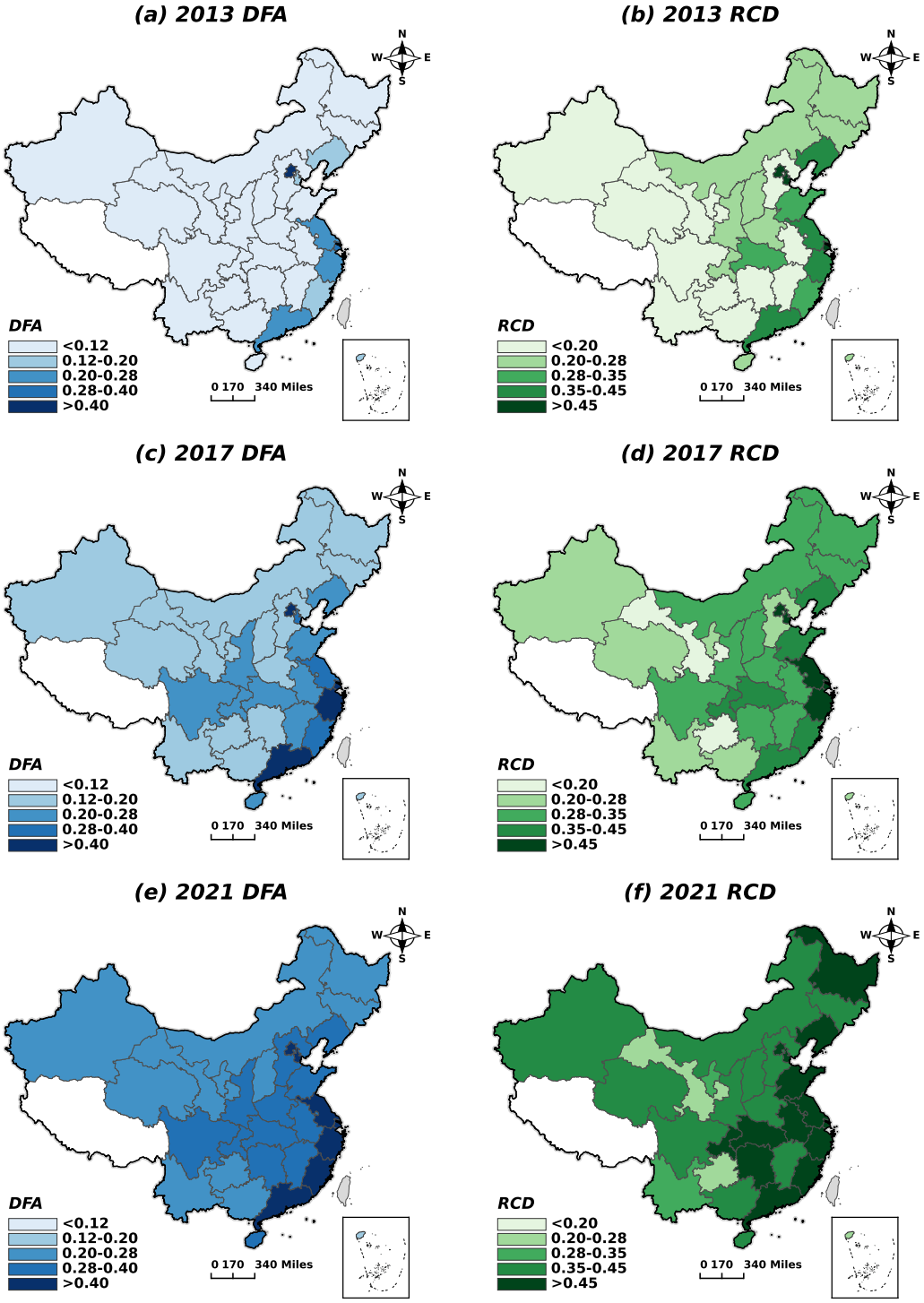


Figure 2. Spatial distribution of DFA (a–c) and RCD (d–f) in 2013, 2017, and 2021. Tibet is shown in white and Taiwan in light gray because they are not included in the estimation sample. The five classes use fixed cutoffs for comparability across years.

Figure 3 combines the cross-section with the time path. Panels (a) and (b) plot every province in 2013 and 2021, with bubble area proportional to GDP per capita on a common scale. The cloud moves up and to the right and the largest bubbles sit in the upper-right corner, so income levels and the two indices move together in the cross-section. The vertical spread also widens: in 2021 Xinjiang and Gansu both have a DFA of about 0.23 but RCD values of 0.372 and 0.260, and Guangdong has a higher DFA than Zhejiang but a lower RCD. Panels (c) and (d) show regional averages. The east keeps a clear lead in both indices;

the west starts lowest but grows fastest in DFA after 2015; the northeast has a relatively high initial RCD and a slow digital transition. The ordering of provinces is stable: the Spearman rank correlation between consecutive years is at least 0.957 for DFA and 0.963 for RCD, and between 2013 and 2021 it is 0.918 and 0.899.

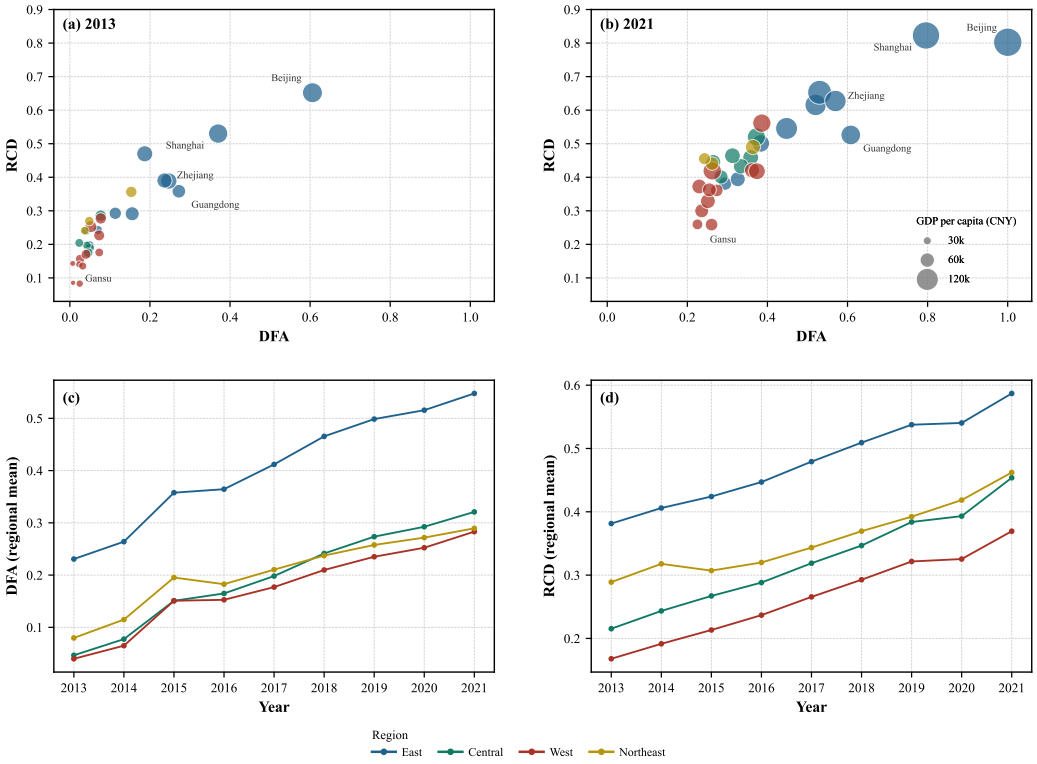


Figure 3. Cross-section and time path of DFA and RCD. (a) DFA against RCD, 2013; (b) DFA against RCD, 2021, with bubble area proportional to GDP per capita on a common scale; (c) regional DFA means; (d) regional RCD means.

Figure 4 shows the full distribution for every year as a density surface. The DFA surface flattens over time: a narrow low-value peak in 2013 gives way to a broad band in 2021, with the mass shifting right and spreading out. The RCD surface shifts right while keeping much of its shape. This is the distributional counterpart of the dispersion measures above.

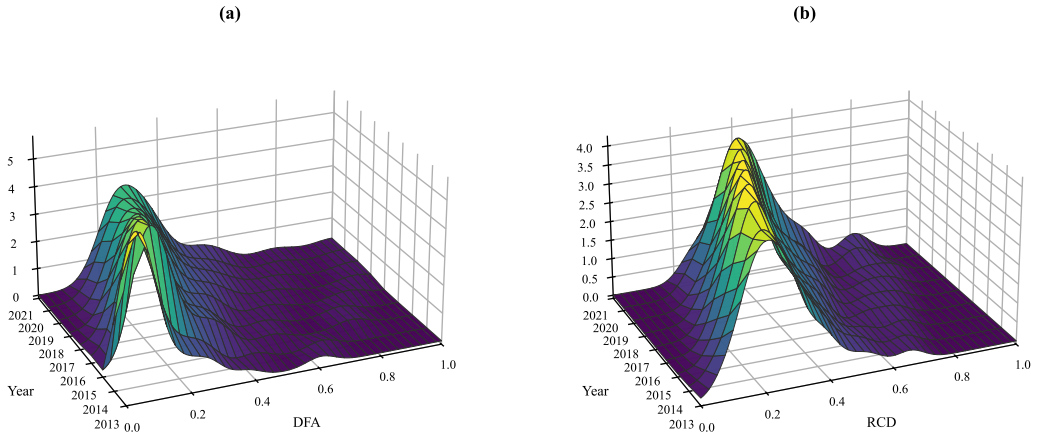


Figure 4. Kernel density surfaces of the DFA and RCD distributions, 2013–2021. (a) DFA; (b) RCD. Densities are evaluated on a common grid of 120 equally spaced index values.

4.2. Baseline Estimates

Table 3 reports the baseline estimates. Column (1) includes DFA with province and year fixed effects and no controls; the coefficient is 0.277 ($p < 0.05$). Adding the controls in column (2) lowers it to 0.241 ($p < 0.05$), and adding region-specific trends in column (3) lowers it to 0.204 ($p < 0.10$). Column (4) uses the five-dimension RCD index as the outcome and returns 0.013, which is not significant.

A one-unit increase in DFA is associated with an increase of 0.241 points in RCD, about 1.6 standard deviations. Since DFA ranges from 0 to 1, this is a large effect at the top of its distribution; a one-standard-deviation increase (0.179) is associated with an increase of about 0.043 points, or 0.29 standard deviations. The controls are individually insignificant, which is expected: they absorb local conditions rather than explain RCD, and their inclusion barely moves the DFA coefficient.

The contrast between columns (2) and (4) shows that DFA is associated with the three core coordination dimensions but not with the extended five-dimension index. The extended index adds infrastructure and ecological coordination, which are slower-moving and may respond to digitalization with a longer lag.

Table 3. Baseline estimates of the effect of DFA on RCD.

	(1)	(2)	(3)	(4)
DFA	0.277**	0.241**	0.204*	0.013
	(0.112)	(0.118)	(0.118)	(0.133)
Government intervention		-0.0012	-0.0007	-0.0001
		(0.001)	(0.001)	(0.001)
Log FDI		0.0215	0.0169	0.0183
		(0.016)	(0.017)	(0.013)
Log population density		0.0794	0.1483	0.1003
		(0.138)	(0.203)	(0.129)
Private economy		-0.00002	0.00011	-0.00016
		(0.000)	(0.000)	(0.000)
Controls	No	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Region trends	No	No	Yes	No
Observations	270	270	270	270

Notes: The dependent variable is RCD in columns (1)–(3) and RCD5 in column (4). DFA is the digital factor allocation index. Controls include government intervention, log FDI per capita, log population density, and private economy vitality. Province and year fixed effects are included in all columns. Region trends are region-specific linear trends. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.3. Threshold Estimates

Table 4 reports the threshold tests and regime coefficients. The economic-development threshold is the main result. The estimated threshold is 10.589 in log GDP per capita, about CNY 39,700 per person, with a bootstrap p-value of 0.003. The low-regime coefficient is 0.506 and the high-regime coefficient is 0.677, both significant at the 1% level, so the marginal association is about 34% larger above the threshold. Because the threshold model absorbs only province fixed effects, these coefficients are not directly comparable to Table 3; the relevant comparison is between regimes. The pattern supports H2.

Figure 5 plots the likelihood-ratio statistic. The minimum is sharp and the 95% confidence interval is narrow, from 10.583 to 10.639. Figure 6 shows the raw association split by the threshold. The high-development regime has the steeper slope, although the

raw scatter also reflects the strong cross-sectional correlation between the two indices, so the threshold is best read together with the fixed-effects estimates.

Table 4. Threshold tests and regime-specific coefficients.

Panel A: Threshold tests

Threshold variable	$\hat{\gamma}$	95% CI	F	Bootstrap p
DFA	0.178	[0.153, 0.349]	15.50	0.137
ln(per capita GDP)	10.589	[10.583, 10.639]	29.07	0.003
Financial development	10.045	–	61.48	0.010
Technological investment	0.856	[0.856, 0.998]	22.70	0.120
Private economy vitality	368.402	[85.480, 368.402]	23.81	0.107

Panel B: Regime-specific coefficients of DFA

Threshold variable	Low regime	High regime	Observations (low/high)
DFA	0.549***	0.661***	90/180
	(0.060)	(0.066)	
ln(per capita GDP)	0.506***	0.677***	66/204
	(0.057)	(0.067)	
Financial development	0.578***	0.267***	240/30
	(0.065)	(0.082)	
Technological investment	0.812***	0.627***	33/237
	(0.088)	(0.051)	
Private economy vitality	0.652***	0.573***	244/26
	(0.066)	(0.070)	

Notes: Each row corresponds to a separate threshold model with province fixed effects and the full control set. The threshold is estimated by grid search over the inner 80% of the threshold variable's distribution. The F statistic tests the null of a single linear regime. The bootstrap p-value is based on 300 cluster wild bootstrap replications. Standard errors clustered by province are in parentheses. For financial development, the likelihood-ratio interval collapses at the grid resolution; it is therefore not reported. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

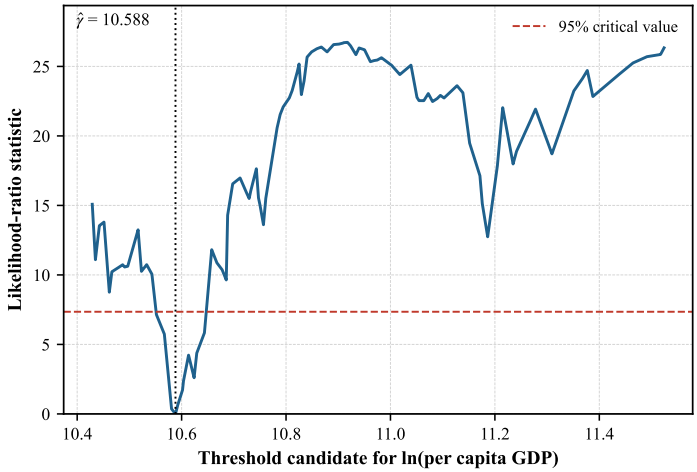


Figure 5. Likelihood-ratio statistic for the economic-development threshold. The dashed line is the 95% critical value (7.35). The dotted vertical line marks the estimated threshold.

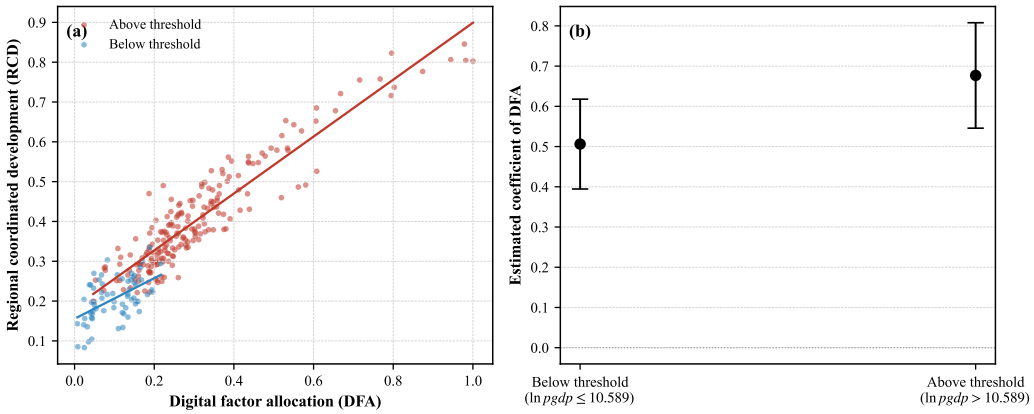


Figure 6. DFA, RCD, and the economic-development threshold. (a) Raw association between DFA and RCD, split by the estimated threshold; the lines are linear fits within each regime. (b) Regime-specific coefficients of DFA with 95% confidence intervals from the threshold model.

The financial-development threshold is also significant. The estimate is 10.045; the low-regime coefficient is 0.578 and the high-regime coefficient 0.267 (bootstrap $p = 0.010$). The high regime contains 30 observations, 27 of them from Beijing, Shanghai, and Tianjin plus three from Qinghai, so the decline is driven by a small group of financially deep municipalities and should not be generalized to all high-finance provinces.

The other threshold variables do not produce significant breaks: the DFA self-threshold has a bootstrap p-value of 0.137, the technology threshold 0.120, and the private-economy threshold 0.107. The nonlinearity is therefore associated with economic development rather than with the level of digitalization itself.

One caveat applies to all threshold estimates. The model follows Hansen (1999) and includes province fixed effects but not year fixed effects, because the threshold variable is a slow-moving development indicator whose cross-province variation year effects would absorb. Adding region-specific linear trends leaves the threshold positive but reduces its precision (bootstrap $p = 0.47$). The threshold is therefore a long-run structural pattern rather than a short-run causal effect.

4.4. Regional Heterogeneity

Under two-way fixed effects the west has a positive and highly significant coefficient of 0.555 (Table 5). The eastern and central coefficients are positive but insignificant (0.227 and 0.192), and the northeast is negative and insignificant (-0.223). Under province fixed effects all four regions are positive and significant, with the largest coefficients in the west (0.535) and centre (0.485), followed by the east (0.296) and the northeast (0.096).

The western result is the most robust: it holds under both specifications and is significant at the 1% level. This supports H4 and the catch-up interpretation, since the marginal return is higher where digital infrastructure and financial access start low. The eastern and central results depend on whether year effects are included, which suggests that their average association partly reflects national trends. The northeastern result is weak and sometimes negative, consistent with the region's industrial legacy and slow digital transition. Subsample estimates are imprecise and the regional differences are not formally tested, so the pattern is descriptive.

Table 5. Regional heterogeneity in the effect of DFA on RCD.

Region	Two-way FE	Province FE
East	0.227	0.296***
	(0.234)	(0.104)
Central	0.192	0.485***
	(0.340)	(0.072)
West	0.555***	0.535***
	(0.125)	(0.077)
Northeast	-0.223	0.096***
	(0.285)	(0.023)
Controls	Yes	Yes
Observations	90/54/99/27	90/54/99/27

Notes: Each cell reports the coefficient of DFA from a separate regression for the indicated region. Controls are included in all columns. Two-way FE includes province and year fixed effects; province FE includes province fixed effects only. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.5. Robustness

Table 6 reports the robustness checks. The coefficient remains positive and significant when the RCD index is replaced by a globally entropy-weighted index (0.349), when DFA is replaced by an equal-weight index (0.298), when municipalities are excluded (0.288), when the sample ends in 2019 (0.283), and when the outcome and regressor are winsorized at the 1st and 99th percentiles (0.302). It is larger under province fixed effects (0.689) and smaller and insignificant for the five-dimension RCD index, the coupling coordination degree, the entropy-weighted DFA index, and lagged DFA.

The pattern is consistent with the baseline contrast. The five-dimension index adds infrastructure and ecological dimensions that respond to DFA slowly, and the coupling coordination degree emphasizes balance among dimensions rather than levels. Dropping 2020 leaves the result intact, so the COVID-19 shock is not driving it. The insignificant lagged coefficient suggests a contemporaneous association, which may reflect the fast diffusion of digital services or the current-state nature of the DFA index.

The entropy-weighted DFA index, which places more weight on IPv4 addresses and domain names, produces a positive but insignificant coefficient. The equal-weight and baseline indices, which spread weight across infrastructure, industry, finance, and human capital, do better, so the result is not an artefact of one indicator. Section 4.8 enters the four dimensions separately.

Figure 7 collects the regional and specification estimates. In panel (a) the western coefficient is the only regional estimate whose interval excludes zero under two-way fixed effects; in panel (b) every specification but the five-dimension RCD index lies entirely above zero.

Table 6. Robustness checks.

Specification	Coefficient	SE	N
Baseline	0.241**	(0.118)	270
Five-dimension RCD	0.013	(0.133)	270
CCD of core dimensions	0.118	(0.288)	270
Entropy-weighted RCD	0.349**	(0.154)	270
Equal-weight DFA	0.298**	(0.143)	270
Entropy-weighted DFA	0.186	(0.176)	270
Excluding municipalities	0.288**	(0.128)	234
2013-2019	0.283**	(0.123)	210
Additional controls	0.213	(0.138)	270
Province FE only	0.689***	(0.069)	270
Lagged DFA	0.139	(0.146)	240
Winsorized 1/99	0.302**	(0.122)	270

Notes: Each row reports the coefficient of DFA (or lagged DFA) from a separate regression. The baseline includes the full control set and two-way fixed effects. The five-dimension RCD index adds infrastructure and ecological coordination. The coupling coordination degree is computed from the three core dimensions. The equal-weight DFA index uses equal weights for all 15 indicators. The entropy-weighted DFA index uses global entropy weights. The additional-controls specification adds technological investment, industrial structure upgrading, and financial development. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

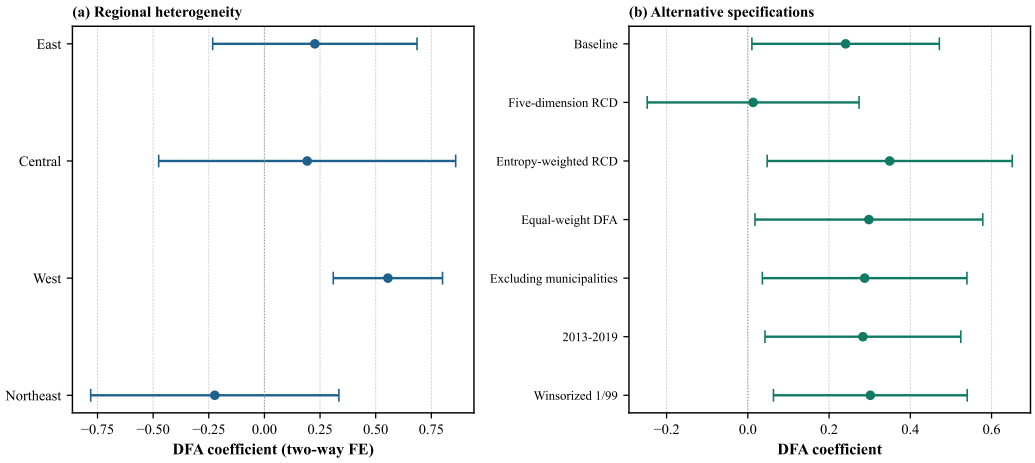


Figure 7. Regional and specification heterogeneity. (a) Regional DFA coefficients under two-way fixed effects. (b) DFA coefficients under alternative specifications. Bars are 95% confidence intervals.

4.6. Mechanisms

DFA is positively associated with technological investment (6.290, $p < 0.05$) and financial development (8.885, $p < 0.01$) under two-way fixed effects, and with industrial structure upgrading under province fixed effects (144.903, $p < 0.01$) (Table 7). These results support the view that DFA operates through innovation, financial deepening, and structural change.

The income-gap result depends on the horizon. Under two-way fixed effects DFA is positively associated with the urban-rural income ratio (0.715, $p < 0.05$), so a one-unit increase widens the gap in the short run; under province fixed effects the association is negative (-1.279, $p < 0.01$), so over the longer run DFA is associated with a narrower gap.

In the short run digital opportunities appear to be captured disproportionately by urban and skilled groups, while in the longer run diffusion to rural areas may narrow the gap.

The mediator coefficients in the RCD equation are not significant, so the channel evidence is suggestive rather than a formal mediation result. The reduced-form effect remains positive when the mediators are included, which suggests that the channels complement rather than fully explain the relationship. A formal mediation analysis would require exogenous variation in each channel; the tests in Appendix C confirm that the indirect effects are not separately identified in either regime.

Table 7. Channel regressions.

Mediator	DFA → M (two-way FE)	DFA → M (province FE)	M in RCD equation
Technological investment	6.290**	1.255	0.006
	(2.985)	(1.275)	
Financial development	8.885***	4.941***	-0.001
	(3.164)	(1.773)	
Industrial upgrading	127.226	144.903***	-0.000
	(80.849)	(29.771)	
Urban-rural income ratio	0.715**	-1.279***	–
	(0.302)	(0.180)	
Controls	Yes	Yes	Yes
Observations	270	270	270

Notes: The first two columns report the effect of DFA on the mediator under two-way and province fixed effects, respectively. The third column reports the coefficient of the mediator in the RCD equation. Controls are included in all columns. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.7. Spatial Dependence

Both indices are spatially clustered (Table 8). Under rook contiguity, Moran's I for DFA ranges from 0.239 to 0.314 and for RCD from 0.377 to 0.427, and the null of no autocorrelation is rejected at the 5% level or better in every year. Under inverse-squared-distance weights the RCD clustering persists (0.117 to 0.175) while the DFA statistic falls to between 0.017 and 0.068 and is only marginally significant in 2013 and 2021. Digital factors therefore concentrate in adjacent provinces—coastal agglomerations and contiguous western blocks—rather than decaying smoothly with distance. The clustering survives the within transformation: after province demeaning, mean Moran's I is 0.323 for DFA and 0.244 for RCD, positive in eight of nine years.

Figure 8 maps the local clusters. Shanghai and Jiangsu are high–high for both indices in both years, Tianjin is high–high for DFA, and Zhejiang joins both clusters by 2021. The low–low block in the west consolidates: for RCD it grows from seven provinces in 2013 to nine in 2021, adding Inner Mongolia, Ningxia, and Shaanxi while Guizhou drops out. Hainan is a low–high outlier throughout. Beijing is high in both indices but is not classified as a cluster because its neighbours are much poorer; it is a single-province peak rather than part of an agglomeration.

Table 8. Global Moran's I for DFA and RCD.

Index	W	Global Moran's I by year							
		2013	2014	2015	2016	2017	2018	2019	2021
DFA	Rook contiguity	0.249**	0.249**	0.240**	0.239**	0.256**	0.269**	0.284**	0.314**
DFA	Inverse distance	0.049*	0.042	0.017	0.029	0.036	0.032	0.041	0.068*
RCD	Rook contiguity	0.413***	0.410***	0.393***	0.377***	0.384***	0.387***	0.377***	0.427***
RCD	Inverse distance	0.175**	0.173***	0.135**	0.124**	0.129**	0.133**	0.117**	0.168**

Notes: Inference is based on 999 random permutations of the cross-section within each year. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Rook contiguity is row-standardized; Hainan is linked to Guangdong because it has no land neighbour. The inverse-distance weights use the squared reciprocal of the distance between provincial centroids.

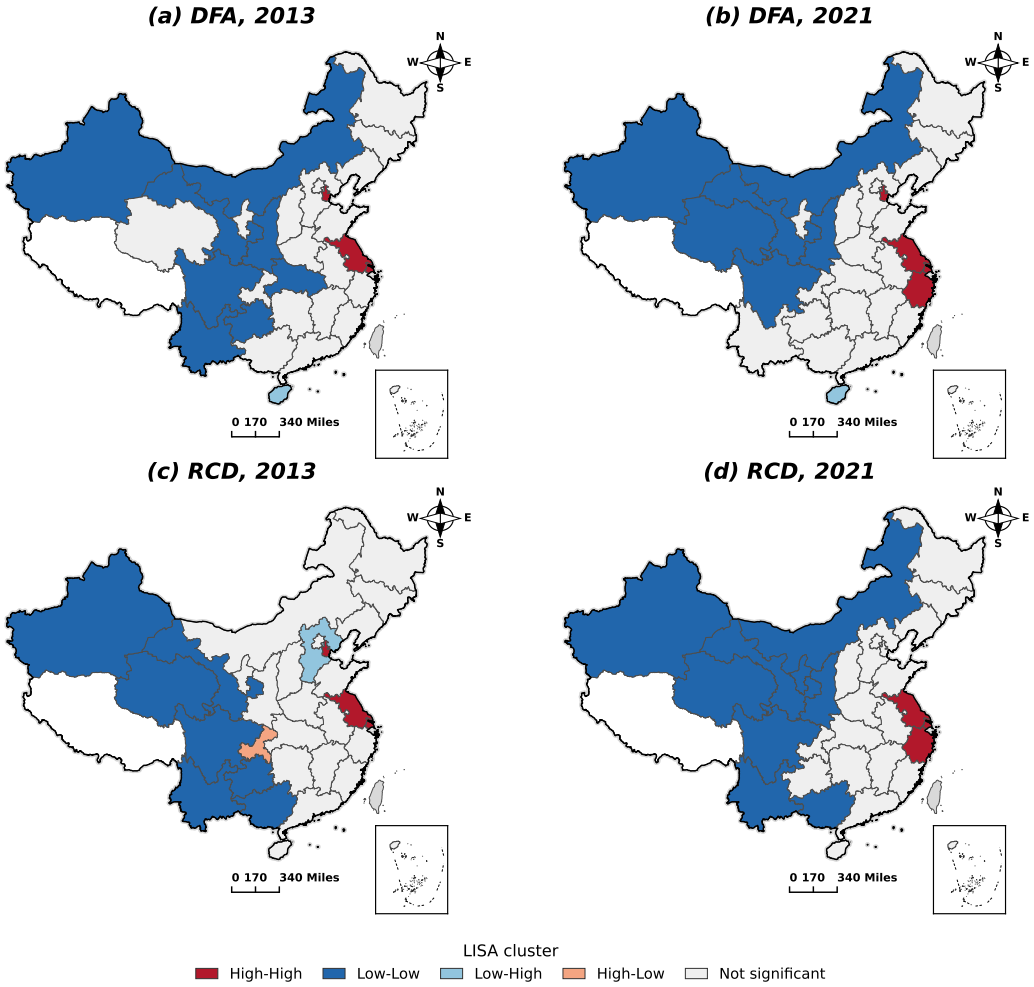


Figure 8. Local Moran clusters (LISA) for DFA and RCD, 2013 and 2021. A province is classified as a cluster when the pseudo p-value from 999 permutations is below 0.10; the remaining provinces are shown as not significant. Tibet is not in the estimation sample and is left blank.

Table 9 asks whether the threshold survives these patterns. The spatial autoregressive specification places the threshold at 10.583, essentially the baseline value of 10.589, and estimates $\rho = 0.580$; the regime coefficients fall to 0.169 and 0.274, the bootstrap p-value for the break rises to 0.143, and the implied total effects are 0.402 and 0.652. The spatial error specification places the threshold at 11.181 (about CNY 71,900) and estimates a declining slope (0.615 against 0.550). The SLX row is reported for completeness only: the coefficient on neighbouring DFA cannot be separated from the own-province coefficient because the two are correlated at 0.979 after the within transformation.

The location of the threshold is therefore stable when the outcome is allowed to depend on neighbouring outcomes, and the positive association holds in both regimes under every specification. The size of the break is not stable, ranging from about 0.17 in the baseline to roughly zero in the spatial error model. We read the threshold as a development gradient rather than a sharp structural break.

Table 9. Threshold estimates with spatial controls.

Model	$\hat{\gamma}$	Bootstrap p	Low regime	High regime	Spatial parameter	N
Baseline	10.589	0.003	0.506*** (0.058)	0.677*** (0.069)		270
SLX[a]	10.589	0.030	0.133 (0.154)	0.275* (0.158)		270
Spatial error	11.181	0.027	0.615*** (0.111)	0.550*** (0.110)	0.720	270
Spatial autoregressive	10.583	0.143	0.169*** (0.055)	0.274*** (0.060)	0.580	270
Panel B: Effects in the SAR model			*Direct*	*Indirect*	*Total*	
Low regime			0.188	0.214	0.402	
High regime			0.305	0.347	0.652	

Notes: All models include province fixed effects and the full control set, and the threshold variable is log GDP per capita. The bootstrap p-value is based on 300 cluster wild bootstrap replications and tests the null of a single linear regime; for the spatial error and spatial autoregressive models the bootstrap is conditional on the estimated spatial parameter. The spatial parameter is λ for the spatial error model and ρ for the spatial autoregressive model. Panel B reports the direct, indirect, and total effects of DFA in the spatial autoregressive model, computed from $(I - \rho W)^{-1}$ with row-standardized weights. The SLX specification (row 2) is not identified. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.8. Which Digital Factor Matters?

Table 10 enters the four normalized dimension scores separately and jointly, and repeats the threshold model for each dimension. Digital finance is the only dimension with an independent association: 0.134 ($p < 0.05$) on its own and 0.151 ($p < 0.01$) when entered jointly with the other three, which are indistinguishable from zero. Digital infrastructure (0.057) and digital industry (0.069) are positive but imprecise, and digital human capital is negative (-0.173) and imprecise. Measured against the three-dimension outcome, financial inclusion is doing much of the work.

The dimensions also have different threshold patterns. Digital infrastructure is positive in both regimes but larger below the threshold (0.587 against 0.278), consistent with diminishing returns to basic connectivity. Digital industry and digital human capital switch from negative below the threshold to positive above it (-0.250 to 0.113 and -0.341 to 0.102), the pattern an absorptive-capacity argument predicts. Digital finance is positive in both regimes and slightly larger above (0.168 to 0.234). The rising pattern of the composite index is the net of these four, which is why it is milder than the individual breaks for industry and human capital.

The negative low-regime coefficients should not be read as evidence that digital industry or digital skills harm lagging provinces. They are more plausibly a failure of these factors to translate into coordination without the complementary conditions that define the high regime. The low-income regime contains only 66 observations, so these slopes are imprecise.

Table 10. Digital factor dimensions: separate, joint, and threshold estimates.

Panel A: Two-way fixed effects, each dimension entered separately and jointly

	Separate		Joint		Threshold model	
Dimension	Coefficient	p	Coefficient	p	Low regime	High regime
Digital infrastructure	0.044	0.539	0.057	0.460	0.587	0.278
Digital industry	0.069	0.385	0.069	0.320	-0.250	0.113
Digital finance	0.134**	0.012	0.151***	0.003	0.168	0.234
Digital human capital	-0.113	0.270	-0.173	0.142	-0.341	0.102

Dimension	$\hat{\gamma}$	F	Bootstrap p	N
Digital infrastructure	11.178	39.70	0.005	270
Digital industry	10.586	47.53	<0.005	270
Digital finance	10.586	28.05	0.005	270
Digital human capital	10.586	61.75	<0.005	270

Notes: Each dimension score is the entropy-weighted within-dimension index, min–max normalized over the panel. In the upper table, columns (1) and (2) report each dimension entered separately, columns (3) and (4) report all four dimensions entered jointly, and columns (5) and (6) report the regime-specific coefficients from a separate threshold model for each dimension. The lower table reports the estimated threshold and the bootstrap p-value for each dimension. All models include province and year fixed effects (threshold models: province fixed effects only) and the full control set. Standard errors clustered by province. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.9. Specification and Identification Checks

Panel A of Table 11 reports Oster (2019) bounds, which ask how strong selection on unobservables would have to be, relative to the observed controls, to drive the DFA coefficient to zero. Under two-way fixed effects the coefficient moves from 0.277 without controls to 0.241 with them, and the bias-adjusted value at $\delta = 1$ is 0.213. The value of δ required to reach zero is 8.58, against the benchmark of 1 that Oster suggests. Under province fixed effects the required value is 27.4; adding the three additional controls lowers the coefficient to 0.213 and the required value to 1.74, still above the benchmark.

Panel B reports a randomisation placebo. If the threshold reflected the true development ordering, reassigning income levels across provinces within each year should destroy it. Over 499 permutations the mean Hansen F is 5.85 and its 95th percentile is 12.00, against an observed value of 26.74 ($p = 0.002$), and the permuted threshold estimates are dispersed (median 10.978, interquartile range 10.604 to 11.276). The threshold is not an artefact of searching over a spatially clustered variable.

A continuous moderation specification points the other way. Interacting DFA with log GDP per capita gives a coefficient of 0.012 ($p = 0.90$), and with financial development -0.002 ($p = 0.75$) (Table A4). A regime shift and a smooth gradient are different hypotheses, and the nonlinearity in this paper rests on the threshold parameterisation.

We do not report an instrumental-variable or dynamic-panel threshold estimate. Such estimators require an external instrument for DFA—historical telephone penetration, terrain ruggedness, or distance to the backbone network—none of which is available here, and we do not construct a pseudo-instrument from the indicators that enter the index. The estimates are therefore conditional associations, robust to the observed confounders and to spatial dependence. The formal mediation test in Appendix C points the same way: the bootstrap intervals for the indirect effects include zero in both regimes.

Table 11. Oster bounds and randomisation inference.

Panel A: Oster (2019) bounds				
Specification	$\tilde{\beta}$	R^2	β^*	δ^*
Two-way FE, baseline controls	0.241	0.171	0.213	8.58
Province FE, baseline controls	0.689	0.895	0.664	27.40
Two-way FE, additional controls	0.213	0.184	0.090	1.74
Panel B: Randomisation inference for the threshold				
Observed F	Mean	95th pct.	p-value	Reps.
26.74	5.85	12.00	0.002	499

Notes: Panel A reports Oster (2019) bounds. $\tilde{\beta}$ is the coefficient of DFA with the full control set and β^* is the bias-adjusted coefficient at $\delta = 1$ with $R_{\max} = 1.3\tilde{R}^2$ (capped at 1). δ^* is the degree of selection on unobservables, relative to selection on observables, that would be required to drive the coefficient to zero. Panel B permutes log GDP per capita across provinces within each year and re-estimates the threshold model 499 times; the p-value is the share of permuted F statistics at least as large as the observed one.

5. Discussion

5.1. Main Findings

This study asked whether digital factor allocation promotes regional coordinated development and whether the relationship is nonlinear. The answer is yes on both counts, with qualifications. On average DFA is positively associated with RCD, and the estimate of 0.241 survives alternative weighting of the outcome, the exclusion of municipalities and of 2020, and winsorization. The association is stronger in provinces that have crossed a critical level of economic development and is concentrated in the west. A financial-development threshold indicates declining returns at very high levels of financial depth, although that result rests on a small group of municipalities.

Three points follow for the literature. First, the digital dividend is conditional: digital inputs are more productive when they can be combined with skills, markets, and institutions. Second, the digital-divide and digital-dividend views are both right at different stages of development: access without absorptive capacity does not convert into coordination gains, and once the capacity is in place the same inputs do more. Third, a composite index conceals divergent behaviour across its components. Digital finance carries most of the association, while industrial digitalization and digital human capital change sign across the threshold.

The spatial analysis narrows the claim. The positive association holds in every specification and in both regimes, and the threshold location is stable when the outcome is allowed to depend on neighbouring outcomes. The size of the gap between regimes is not stable. The threshold therefore describes a development gradient in the cross-section, not a sharp structural boundary.

5.2. Absorptive Capacity and the Digital Divide

The economic-development threshold is consistent with absorptive capacity. Digital technologies are general-purpose technologies whose effects depend on organizational change, human capital, and complementary infrastructure (Brynjolfsson & McAfee, 2014; O'Mahony & Vecchi, 2005). In low-income provinces digital infrastructure may be used mainly for communication and consumption, while productive and coordination effects require inputs that are in short supply. In higher-income provinces digital inputs can be combined with skilled labour, financial services, and market institutions.

This is also consistent with the multi-level digital divide. Access is the first level, skills the second, and the ability to convert access into material benefits the third (Norris, 2001; van Dijk, 2005; Warschauer, 2003). Our results suggest that the third-level divide binds in low-income provinces, so providing access alone is not enough.

The regional and threshold results are not in conflict. The regional estimate compares average outcomes across regions over the sample period, while the threshold estimate compares provinces at different development stages within a period. The west can have a larger average association because it starts from a lower base even if the marginal association at a given development level is smaller.

5.3. Financial Development and Diminishing Returns

The financial-development threshold suggests that the marginal association of DFA declines at very high levels of financial depth, consistent with diminishing returns to digital finance at the extensive margin. Where financial systems are shallow, digital finance expands access to basic services; where they are deep, it adds less to access and may instead intensify competition or encourage speculative activity (Beck et al., 2007). The result rests on 30 high-regime observations, mostly Beijing, Shanghai, and Tianjin, and should be read as a caution against assuming that more digital finance always produces more coordination.

The decomposition of the index points at the same factor from another direction. Entered jointly, digital finance is the only dimension with an independent association with RCD, and its regime pattern mirrors the composite: positive in both regimes and slightly larger above the threshold. Financial depth therefore matters twice, as a moderator of the association and as the carrier of most of it. The two roles are conceptually distinct but point to the same policy margin: the quality and direction of digital finance rather than its volume.

5.4. Regional Heterogeneity

The association is strongest in the west and weaker or insignificant elsewhere under two-way fixed effects, which supports the inclusive-digitalization view: where digital infrastructure and financial access are scarce, additional digital factors have larger marginal effects. The northeast is weaker and sometimes negative. It has relatively high initial urbanization and public services but a slow digital transition and a declining working-age population, so digital factor allocation may not offset these structural constraints without a broader revitalization strategy.

5.5. Policy Implications

Digital factor allocation should be sequenced with economic development. In low-income provinces the priority is basic connectivity, digital literacy, and complementary public services; advanced digital industry and digital finance are unlikely to raise coordination until those foundations are in place. In higher-income provinces the priority shifts to deeper integration of digital technologies into production, public services, and urban-rural linkages.

Policies should be tailored to regional absorptive capacity. The west has the strongest average association and should be a priority for digital infrastructure and digital finance investment, paired with human capital development and market reform. Because the short-run association with the urban-rural income ratio is positive, gains may be captured disproportionately by urban and skilled groups; rural e-commerce, digital agriculture, digital public services, and targeted skills training can broaden them. Advanced regions should attend to the quality and direction of digital investment rather than its quantity, given the declining association at high financial depth.

5.6. Limitations

Four limitations deserve emphasis. First, the analysis covers 30 provinces and does not capture within-province heterogeneity; city-level data would provide a more granular picture. Second, the indices depend on indicator selection and weighting. We use

alternative weighting schemes and alternative outcome definitions, but the indices remain simplifications.

Third, the estimates are conditional associations. DFA may be endogenous to RCD, and the lagged specification is not significant, which suggests a contemporaneous relationship that could reflect reverse causality. Oster bounds and the randomisation placebo indicate that unobservables would have to be more important than the observed controls to eliminate the association, but they do not identify a causal effect. External instruments such as historical telephone penetration, terrain ruggedness, or distance to the backbone network are not available for this sample, and we do not construct a pseudo-instrument from the index components. Future work should exploit such variation.

Fourth, the threshold is sensitive to specification. Region-specific trends, year fixed effects, and spatial specifications either reduce its precision or move its location, and the size of the break is not stable across spatial models. The threshold should be read as a long-run pattern, not a short-run causal effect. The sub-dimension slopes are also imprecise because the low-income regime contains 66 observations, and the income series carry measurement error from the household survey and from the urban-rural classification. Finally, the sample includes the COVID-19 period; the results are robust to dropping 2020, but 2021 may still be affected.

6. Conclusions

This study constructed a four-dimension DFA index covering digital infrastructure, digital industry, digital finance, and digital human capital, and a three-dimension RCD index covering economic, social, and urban-rural coordination, for 30 Chinese provinces from 2013 to 2021. Using a Hansen panel threshold model, it tested whether the effect of DFA on RCD depends on economic development and financial development, and it examined which dimension carries the association and whether the estimates survive spatial dependence.

DFA is positively associated with RCD. The baseline two-way fixed-effects estimate is 0.241 and the association holds across alternative weighting, sample restrictions, and winsorization. It is economically meaningful: a one-standard-deviation increase in DFA is associated with an increase of about 0.043 points in RCD, or 0.29 standard deviations.

The association rises with economic development. Above the estimated threshold of 10.589 in log GDP per capita (about CNY 39,700), the marginal association is 0.677 against 0.506 below, consistent with absorptive capacity. A financial-development threshold indicates declining returns at very high levels of financial depth, driven by a small group of financially deep municipalities. Both indices are spatially clustered; allowing for spatial dependence leaves the threshold location unchanged, so we read it as a development gradient rather than a sharp break.

Two findings extend the usual account. Within the composite, digital finance is the only dimension with an independent association, and it also shows the most stable positive slope across regimes. Industrial digitalization and digital human capital, by contrast, switch from negative below the threshold to positive above it, which is what an absorptive-capacity argument predicts. The regional analysis shows the association concentrated in the west, where digital infrastructure and financial access start low.

These findings imply a differentiated policy. Digital factor allocation should be sequenced with economic development: basic connectivity and digital literacy first in low-income provinces, deeper integration of digital technologies into production and public services in higher-income ones. The west should be a priority for digital infrastructure and digital finance investment, paired with human capital development and market reform. Because the short-run association with the urban-rural income ratio is positive, comple-

mentary rural digital services and targeted skills training are needed to share the gains. Advanced regions should focus on the quality and direction of digital investment rather than its quantity.

Several limitations remain. The analysis uses provincial data and does not capture within-province heterogeneity; the indices depend on indicator and weighting choices; and the estimates are conditional associations rather than identified causal effects. The threshold is sensitive to the treatment of trends, year effects, and spatial dependence. Future work can address these points with city-level data, external instruments, dynamic spatial threshold models, and a DFA index extended to data governance, digital security, and the quality of digital public services.

Digitalization is not a universal remedy for regional disparities. Its payoff depends on the stage of economic development, the depth of the financial system, and regional absorptive capacity, and a differentiated approach that combines digital investment with complementary economic and social reforms is more likely to deliver inclusive coordination gains.

Author Contributions: Conceptualization, J.W. (Junfeng Wang) and Y.W.; methodology, J.W. (Junfeng Wang); formal analysis, J.W. (Junfeng Wang); data curation, J.W. (Jiayi Wang); writing—original draft, J.W. (Junfeng Wang); writing—review and editing, J.W. (Jiayi Wang) and Y.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The provincial panel data are compiled from publicly available official yearbooks and the Peking University Digital Financial Inclusion Index. The analysis panel, the index construction scripts, and the replication code are available at https://github.com/KarlHeinrich-jpg/Digital-Factor-Allocation-and-Regional-Coordinated-Development.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A Index Weights

Table A1 reports the within-dimension entropy weights for DFA and the global entropy weights for RCD. The DFA index assigns equal weights to the four dimensions. The RCD baseline assigns equal weights to the three dimensions; the weights below refer to the global entropy-weighted alternative used in the robustness checks.

Table A1. Entropy weights for DFA and RCD indicators.

Dimension	Indicator	Weight
Panel A: DFA (equal weights across four dimensions)		
Digital infrastructure	Internet access ports per capita	0.057
Digital infrastructure	Optical cable density	0.211
Digital infrastructure	Mobile phone penetration	0.056
Digital infrastructure	IPv4 addresses per capita	0.414
Digital infrastructure	Domain names per capita	0.262
Digital industry	Software revenue/GDP	0.336
Digital industry	E-commerce sales/GDP	0.214
Digital industry	Share of firms with e-commerce	0.085
Digital industry	Websites per 100 firms	0.042
Digital industry	Information firms per capita	0.323
Digital finance	DFI coverage breadth	0.303
Digital finance	DFI usage depth	0.376
Digital finance	DFI digitization	0.322
Digital human capital	ICT employment share	0.762
Digital human capital	Internet penetration	0.238
Panel B: RCD (global entropy weights)		
Economic	GDP per capita	0.157
Economic	Retail sales per capita	0.156
Economic	Fiscal self-sufficiency	0.084
Social	Disposable income per capita	0.150
Social	Hospital beds per 10,000	0.080
Social	Higher-education students per 100,000	0.073
Social	Social security spending share	0.102
Urban-rural	Urbanization rate	0.081
Urban-rural	Urban-rural income ratio (negative)	0.049
Urban-rural	Urban unemployment rate (negative)	0.066

Notes: Panel A reports within-dimension entropy weights for DFA; each of the four dimensions receives a weight of 0.25 in the composite. Panel B reports global entropy weights for the ten RCD indicators, used in the entropy-weighted RCD robustness check.

Appendix B Index Values by Province and Year

Tables A2 and A3 report the province-by-year values of the two composite indices. Provinces are grouped by region in the order used throughout the paper. The final columns report the province-level mean, the standard deviation over 2013–2021, and the change between 2013 and 2021; the bottom row reports the corresponding national figures. All values are reported to three decimal places.

Table A2. Digital factor allocation (DFA) index values by province and year.

Province	2013	2014	2015	2016	2017	2018	2019	2020	2021	Mean	SD	Change
Eastern												
Beijing	0.606	0.655	0.795	0.803	0.874	0.944	0.978	0.982	1.000	0.849	0.145	+0.394
Fujian	0.156	0.182	0.279	0.298	0.375	0.415	0.434	0.412	0.448	0.333	0.110	+0.292
Guangdong	0.272	0.301	0.380	0.390	0.439	0.519	0.562	0.580	0.608	0.450	0.123	+0.336
Hainan	0.070	0.107	0.217	0.193	0.233	0.285	0.312	0.303	0.326	0.227	0.091	+0.256
Hebei	0.049	0.068	0.137	0.150	0.187	0.219	0.246	0.261	0.294	0.179	0.085	+0.245
Jiangsu	0.247	0.269	0.348	0.347	0.388	0.437	0.471	0.493	0.530	0.392	0.099	+0.283
Shandong	0.113	0.140	0.211	0.231	0.261	0.311	0.322	0.342	0.383	0.257	0.092	+0.270
Shanghai	0.370	0.436	0.534	0.550	0.607	0.668	0.716	0.767	0.796	0.605	0.146	+0.426
Tianjin	0.187	0.223	0.319	0.319	0.343	0.394	0.437	0.479	0.520	0.358	0.111	+0.333
Zhejiang	0.236	0.259	0.357	0.363	0.411	0.461	0.508	0.536	0.570	0.411	0.118	+0.334
Central												
Anhui	0.050	0.096	0.168	0.187	0.223	0.268	0.302	0.322	0.358	0.219	0.104	+0.308
Henan	0.024	0.048	0.119	0.138	0.170	0.211	0.240	0.257	0.284	0.166	0.092	+0.260
Hubei	0.077	0.109	0.189	0.204	0.242	0.290	0.337	0.350	0.372	0.241	0.106	+0.295
Hunan	0.046	0.069	0.141	0.152	0.184	0.226	0.260	0.285	0.313	0.186	0.093	+0.267
Jiangxi	0.042	0.082	0.154	0.160	0.204	0.246	0.278	0.303	0.334	0.200	0.100	+0.292
Shanxi	0.041	0.061	0.134	0.147	0.165	0.207	0.223	0.238	0.264	0.165	0.077	+0.224
Western												
Chongqing	0.077	0.116	0.195	0.220	0.256	0.297	0.327	0.358	0.386	0.248	0.106	+0.309
Gansu	0.007	0.043	0.120	0.099	0.128	0.159	0.183	0.196	0.225	0.129	0.072	+0.218
Guangxi	0.023	0.044	0.120	0.133	0.150	0.193	0.222	0.248	0.273	0.156	0.087	+0.250
Guizhou	0.008	0.035	0.110	0.134	0.156	0.188	0.209	0.219	0.260	0.147	0.084	+0.252
Inner Mongolia	0.052	0.075	0.161	0.173	0.179	0.188	0.209	0.225	0.262	0.169	0.067	+0.210
Ningxia	0.026	0.054	0.149	0.123	0.161	0.188	0.200	0.213	0.251	0.152	0.074	+0.226
Qinghai	0.032	0.045	0.162	0.153	0.157	0.186	0.203	0.216	0.255	0.157	0.074	+0.223
Shaanxi	0.073	0.113	0.192	0.209	0.236	0.284	0.328	0.348	0.374	0.240	0.104	+0.301
Sichuan	0.073	0.099	0.194	0.207	0.245	0.281	0.316	0.337	0.362	0.235	0.101	+0.288
Xinjiang	0.040	0.047	0.134	0.105	0.134	0.166	0.186	0.198	0.230	0.138	0.065	+0.189
Yunnan	0.025	0.043	0.121	0.124	0.143	0.176	0.203	0.218	0.236	0.143	0.074	+0.211
Northeastern												
Heilongjiang	0.037	0.068	0.141	0.143	0.170	0.189	0.216	0.232	0.243	0.160	0.071	+0.206
Jilin	0.049	0.082	0.166	0.161	0.193	0.227	0.232	0.240	0.262	0.179	0.073	+0.213
Liaoning	0.153	0.194	0.279	0.243	0.268	0.296	0.325	0.343	0.364	0.274	0.069	+0.210
National mean	0.109	0.139	0.224	0.229	0.263	0.304	0.333	0.350	0.379	0.259	0.179	+0.271

Notes: Values are the composite DFA index defined in Section 3. Mean and SD refer to the province's own 2013–2021 series; Change is the difference between 2021 and 2013. The national row reports the unweighted mean across the 270 province-year observations.

Table A3. Regional coordinated development (RCD) index values by province and year.

Province	2013	2014	2015	2016	2017	2018	2019	2020	2021	Mean	SD	Change
Eastern												
Beijing	0.652	0.678	0.716	0.737	0.777	0.806	0.846	0.805	0.802	0.758	0.065	+0.150
Fujian	0.291	0.322	0.341	0.352	0.380	0.428	0.469	0.479	0.545	0.401	0.085	+0.254
Guangdong	0.358	0.381	0.382	0.407	0.431	0.460	0.487	0.492	0.526	0.436	0.058	+0.168
Hainan	0.243	0.266	0.294	0.301	0.320	0.335	0.354	0.363	0.394	0.319	0.048	+0.151
Hebei	0.198	0.222	0.229	0.246	0.274	0.307	0.333	0.336	0.381	0.281	0.062	+0.183
Jiangsu	0.389	0.419	0.449	0.479	0.512	0.547	0.572	0.584	0.653	0.512	0.086	+0.264
Shandong	0.293	0.316	0.321	0.341	0.370	0.388	0.417	0.444	0.501	0.377	0.068	+0.208
Shanghai	0.530	0.550	0.584	0.643	0.685	0.721	0.755	0.758	0.823	0.672	0.102	+0.292
Tianjin	0.470	0.490	0.487	0.495	0.528	0.552	0.563	0.564	0.616	0.529	0.048	+0.146
Zhejiang	0.390	0.416	0.437	0.469	0.515	0.548	0.579	0.579	0.627	0.507	0.083	+0.237
Central												
Anhui	0.190	0.219	0.253	0.273	0.307	0.342	0.374	0.408	0.459	0.314	0.090	+0.269
Henan	0.204	0.232	0.256	0.278	0.308	0.330	0.352	0.371	0.401	0.304	0.066	+0.197
Hubei	0.286	0.332	0.371	0.403	0.425	0.452	0.483	0.436	0.520	0.412	0.073	+0.234
Hunan	0.175	0.207	0.233	0.253	0.288	0.327	0.387	0.414	0.464	0.305	0.099	+0.289
Jiangxi	0.197	0.216	0.240	0.263	0.291	0.313	0.354	0.371	0.433	0.298	0.078	+0.236
Shanxi	0.241	0.253	0.250	0.260	0.293	0.315	0.353	0.359	0.445	0.308	0.068	+0.204
Western												
Chongqing	0.277	0.310	0.343	0.372	0.411	0.453	0.498	0.452	0.561	0.409	0.092	+0.284
Gansu	0.143	0.158	0.171	0.184	0.183	0.199	0.216	0.223	0.260	0.193	0.036	+0.116
Guangxi	0.140	0.167	0.192	0.212	0.264	0.287	0.299	0.312	0.361	0.248	0.074	+0.220
Guizhou	0.085	0.098	0.131	0.160	0.199	0.222	0.247	0.249	0.259	0.183	0.067	+0.173
Inner Mongolia	0.253	0.283	0.277	0.291	0.316	0.336	0.353	0.375	0.419	0.323	0.053	+0.166
Ningxia	0.156	0.181	0.208	0.228	0.250	0.270	0.296	0.302	0.329	0.247	0.058	+0.173
Qinghai	0.135	0.156	0.174	0.193	0.221	0.247	0.301	0.324	0.363	0.235	0.080	+0.227
Shaanxi	0.227	0.252	0.272	0.292	0.319	0.349	0.374	0.379	0.418	0.320	0.064	+0.191
Sichuan	0.176	0.199	0.228	0.252	0.290	0.334	0.369	0.379	0.421	0.294	0.086	+0.245
Xinjiang	0.170	0.200	0.217	0.251	0.266	0.298	0.329	0.322	0.372	0.269	0.067	+0.202
Yunnan	0.083	0.104	0.134	0.168	0.203	0.225	0.256	0.262	0.300	0.193	0.075	+0.217
Northeastern												
Heilongjiang	0.241	0.266	0.253	0.270	0.300	0.335	0.374	0.412	0.455	0.323	0.076	+0.215
Jilin	0.270	0.303	0.285	0.294	0.307	0.335	0.367	0.392	0.441	0.333	0.057	+0.171
Liaoning	0.356	0.384	0.383	0.396	0.423	0.438	0.436	0.451	0.490	0.418	0.041	+0.134
National mean	0.261	0.286	0.304	0.325	0.355	0.383	0.413	0.420	0.468	0.357	0.148	+0.207

Notes: Values are the baseline three-dimension RCD index defined in Section 3. Mean and SD refer to the province's own 2013–2021 series; Change is the difference between 2021 and 2013. The national row reports the unweighted mean across the 270 province-year observations.

Appendix C Additional Mechanism and Specification Tables

Table A4 reports continuous moderation specifications, in which DFA is interacted with a centered moderator and the marginal effect of DFA is evaluated at the quartiles of the moderator. Table A5 reports regime-specific mediation tests, estimated separately below and above the economic-development threshold of 10.589. The indirect effect is the product of the DFA-to-mediator path and the mediator-to-RCD path; the confidence interval is obtained from a province-level bootstrap with 300 replications.

Table A4. Continuous moderation of the DFA–RCD association.

	Coefficient	SE	p
Moderator: ln *GDP per capita*			
DFA	0.097	0.137	0.481
DFA × moderator	0.012	0.098	0.904
Marginal effect, p25	0.093	0.156	0.552
Marginal effect, p50	0.096	0.141	0.496
Marginal effect, p75	0.099	0.124	0.423
Moderator: financial development			
DFA	0.262*	0.149	0.080
DFA × moderator	-0.002	0.007	0.749
Marginal effect, p25	0.266*	0.154	0.084
Marginal effect, p50	0.264*	0.151	0.080
Marginal effect, p75	0.261*	0.148	0.077

Notes: The dependent variable is RCD. All models include the DFA index, the centered moderator, their interaction, the four baseline controls, and province and year fixed effects. Marginal effects are evaluated at the quartiles of the moderator and their standard errors are computed from the full covariance matrix of the interaction model. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A5. Regime-specific mediation tests.

Mediator	Regime	a	b	$a \times b$	95% CI	N
Technological investment	Low	3.671	0.0255	0.0935	[-0.043, 0.272]	66
Technological investment	High	3.010	0.0014	0.0043	[-0.027, 0.112]	204
Financial development	Low	5.910	-0.0070	-0.0416	[-0.262, 0.188]	66
Financial development	High	8.676	-0.0013	-0.0109	[-0.226, 0.081]	204
Industrial upgrading	Low	-143.049	-0.0001	0.0084	[-0.180, 0.220]	66
Industrial upgrading	High	56.461	-0.0001	-0.0041	[-0.042, 0.132]	204

Notes: a is the coefficient of DFA in the mediator equation, b is the coefficient of the mediator in the RCD equation, and $a \times b$ is the indirect effect. Both equations include the four baseline controls and province and year fixed effects. Regimes are defined by the estimated threshold of 10.589 in log GDP per capita. The 95% confidence interval is the percentile interval from a province-level bootstrap with 300 replications. N is the number of province-year observations in the regime.

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